F79SP Session 8: Introduction to Markov Chains

1 Markov chains

The notion of Markov chains is defined as follows.

Definition 1.1. *Consider a* ***discrete-time*** *stochastic process* $\{X_n\}_{n\geq 0}$.

- *Let* S *be a* ***countable state space****.*
- *Suppose that each* X_i *takes values in* S.

Then $\{X_n\}_{n\geq 0}$ *is called a* ***Markov chain*** *if:*

$$\boldsymbol{P}(X_{n+1} = j \,|\, X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = \boldsymbol{P}(X_{n+1} = j \,|\, X_n = i)$$

for all $n \geq 0$ *and* $j, i, i_{n-1}, ..., i_0 \in S$*. This property is called the* ***Markov property****.*

By induction, we can show that Markov property implies:

$$\mathbf{P}(X_{n+m} = j \,|\, X_n = i, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = \mathbf{P}(X_{n+m} = j \,|\, X_n = i)$$

Formulate in terms of future, present(now) and history events, i.e., F, N, H respectively, we have:

$$\mathbf{P}(F \,|\, NH) = \mathbf{P}(F \,|\, N)$$

The Markov property allow us to effectively visualise certain Markov chains as transition/state-space diagrams.

Example 1.1. *Let* $S = \{0, 1\}$ *and consider a Markov chain* $\{X_n\}_{n\geq 0}$ *on* S*:*

- *If* n *is odd,* $P(X_{n+1} = 1|X_n = 1) = P(X_{n+1} = 0|X_n = 0) = 1$.
- *If* n *is even,* $P(X_{n+1} = 1|X_n = 1) = P(X_{n+1} = 0|X_n = 0) = \frac{1}{2}$.

Time-homogeneity

A Markov chain is said to be time-homogeneous if

$$\mathbf{P}(X_{n+1} = j \,|\, X_n = i) = p_{ij} \text{ (independent of } n)$$

for all $n \geq 0$ and $i, j \in S$.

A time-homogeneous Markov chain (THMC) can thus be represented by a single transition diagram, and also in matrix form, called the **transition matrix**:

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & \cdots \\ p_{21} & p_{22} & p_{23} & \cdots \\ p_{31} & p_{32} & p_{33} & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

We can also write $P = (p_{ij})_{i,j \in \mathcal{S}}$. Note that necessarily:

$$p_{ij} \geq 0 \text{ for all } i, j \in \mathcal{S}, \qquad \sum_{j \in \mathcal{S}} p_{ij} = 1 \text{ for all } i \in \mathcal{S},$$

and we say that P satisfies the conditions of being a **stochastic matrix**.

Throughout Chapter 4, **all Markov chains are assumed to be time-homogeneous.**

Some examples of THMC

1. Bonus-malus system (see notes)
2. Health-insurance system (see notes)
3. Random walk on countable state space, and its modified processes.

Example 1.2. *Random walks on countable state space (e.g., $\mathbb{Z}$) are THMC. Let:*

$$S_0 = a, S_n = a + \sum_{i=1}^{n} \xi_i, \ n \geq 1$$

, for i.i.d.r.v.'s ξ_i's. Note that:

$$\begin{aligned} &\boldsymbol{P}(S_{n+1} = j \,|\, S_n = i, S_{n-1} = i_{n-1}, ..., S_0 = i_0) \\ &= \boldsymbol{P}(S_{n+1} - S_n = j - i \,|\, S_n = i, S_{n-1} = i_{n-1}, ..., S_0 = i_0) \\ &= \boldsymbol{P}(\xi_{n+1} = j - i \,|\, S_n = i, S_{n-1} = i_{n-1}, ..., S_0 = i_0) \\ &= \boldsymbol{P}(\xi_{n+1} = j - i) \\ &= \boldsymbol{P}(S_{n+1} = j \,|\, S_n = i) \end{aligned}$$

- *Hence, the RW satisfies the Markov property.*
- *The RW possesses time-homogeneous property as the argument above works too by replacing n with $n+m$, and we know $\boldsymbol{P}(\xi_{n+1} = j - i) = \boldsymbol{P}(\xi_{n+m} = j - i)$ as increments are i.i.d.. Therefore, $\boldsymbol{P}(S_{n+1} = j \,|\, S_n = i) = \boldsymbol{P}(S_{n+m+1} = j \,|\, S_{n+m} = i)$ for every m.*
- *The RW possesses space-homogeneous property since $\boldsymbol{P}(S_{n+1} = j \,|\, S_n = i) = \boldsymbol{P}(\xi_{n+1} = j - i)$ implies that the probability is only dependent on the difference of two states involved, $j - i$, but not the states themselves.*

Transition diagram of SRW is as follows:

Example 1.3 (Random walks with an absorbing barrier)**.** *Consider a SRW with $a > 0$, but now that when this process hits state 0 it remains in that state for all future times.*

- *We say 0 is an absorbing state/barrier.*
- *This is a THMC.*

Example 1.4 (Random walks with a reflecting barrier)**.** *Consider a SRW with $a > 0$, but now whenever the process is in state 0, its next step is to remain in state 0 with probability q or go to state 1 with probability p. This process is again a THMC.*

Probability calculation

Consider a THMC $\{X_n\}_{n\geq 0}$ on some countable state space S with transition matrix P. We first define the notion of (probability) distribution.

Definition 1.2. *A row vector $\pi = (\pi_i)_{i\in S}$ on S is said to be a distribution if $\sum_{i\in S} \pi_i = 1$.*

Define the n-step probabilities as $p_{ij}^{(n)} = \boldsymbol{P}(X_{n+m} = j | X_m = i)$ for $i, j \in S$ and any m (independent of m by time homogeneity).

How can we obtain these probabilities?

- *For $n = 0$, we have $p_{ij}^{(0)} = \boldsymbol{P}(X_{0+m} = j | X_m = i)$.*

- *For $n = 1$, we have $p_{ij}^{(1)} = \boldsymbol{P}(X_{m+1} = j | X_m = i)$.*

- *For $n \geq 2$, we have $p_{ij}^{(n)} = (P^n)_{ij}$. We will provide justification on this during next session!*

Example 1.5. *Let $\{X_n\}_{n\geq 0}$ be a THMC with $S = \{1, 2, 3\}$ with transition matrix $P = (p_{ij})_{i,j} \in S$.*

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

- *Determine $\boldsymbol{P}(X_2 = 3|X_0 = 2)$.*
- *Determine $\boldsymbol{P}(X_5 = 3|X_3 = 2)$.*

Solution:

***Note**: We can use the n-step probabilities from P^n to deduce the probability distribution of X_n, if the initial distribution is given, as follows.*

Result 1.1. *Consider a THMC $\{X_n\}_{n\geq 0}$ on some state space S with transition matrix P. Given an initial distribution $\mu = (\mu_i)_{i\in S} = (P(X_0 = i))_{i\in S}$. Then, the vector μP^n gives the probability distribution of X_n.*

$$\begin{aligned}
\boldsymbol{P}(X_n = j) &= \sum_{k\in S} \boldsymbol{P}(X_n = j, X_0 = k) \\
&= \sum_{k\in S} \boldsymbol{P}(X_n = j | X_0 = k)\boldsymbol{P}(X_0 = k) \\
&= \sum_{k\in S} (P^n)_{kj}\boldsymbol{P}(X_0 = k) \\
&= \sum_{k\in S} (P^n)_{kj}\mu_k \\
&= \sum_{k\in S} \mu_k (P^n)_{kj} \\
&= (\mu P^n)_j
\end{aligned}$$

In short, $\boldsymbol{P}(X_n = j)$ is the j-th component of the row vector μP^n.

Example 1.6. *Let $\{X_n\}_{n\geq 0}$ be a THMC with $S = \{1, 2, 3\}$ with transition matrix $P = (p_{ij})_{i,j} \in S$.*

$$P = \begin{pmatrix} 1 & 0 & 0 \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

Given the initial distribution: $\mu = (\boldsymbol{P}(X_0 = 1), \boldsymbol{P}(X_0 = 2), \boldsymbol{P}(X_0) = 3) = (0, \frac{1}{3}, \frac{2}{3})$.

- *Calculate $\boldsymbol{P}(X_2 = 1|X_0 = 2)$.*
- *Calculate $\boldsymbol{P}(X_2 = 1)$.*
- *Calculate $\boldsymbol{P}(X_0 = 2|X_2 = 1)$. (Note: $\boldsymbol{P}(A|B) \neq \boldsymbol{P}(B|A)$ in general)*

Solution: Compute $P^2 = \begin{pmatrix} 1 & 0 & 0 \\ \frac{2}{3} & \frac{7}{36} & \frac{5}{36} \\ \frac{1}{4} & \frac{5}{12} & \frac{1}{3} \end{pmatrix}$. Then:

Chapman-Kolmogorov equations

Let $\{X_n\}_{n\geq 0}$ be a THMC on countable state space S. We have the ***Chapman-Kolmogorov equations*** *as below. For any $m > 0$, $n > 0$, and $i, j \in \mathcal{S}$:*

$$p_{ij}^{(m+n)} = \sum_{k\in\mathcal{S}} p_{ik}^{(n)} p_{kj}^{(m)}.$$

Proof.

$$\begin{aligned} p_{ij}^{(m+n)} &= \mathbf{P}(X_{m+n} = j \,|\, X_0 = i) \\ &= \sum_{k\in\mathcal{S}} \mathbf{P}(X_n = k,\, X_{m+n} = j \,|\, X_0 = i) \end{aligned}$$

and the ***Chapman-Kolmogorov inequalities****: For any $m > 0$, $n > 0$, and $i, j, k \in \mathcal{S}$,*

$$p_{ij}^{(m+n)} \geq p_{ik}^{(m)} p_{kj}^{(n)},$$

which follow immediately (as the summation consists of only non-negative terms).

Remark 1.1. *In particular, when $m = n = 1$, the Chapman-Kolmogorov equations become:*

$$\begin{aligned} p_{ij}^{(2)} &= \sum_{k\in\mathcal{S}} p_{ik}^{(1)} p_{kj}^{(1)} \\ &= \sum_{k\in\mathcal{S}} P_{ik} P_{kj} \\ &= (P^2)_{ij} \end{aligned}$$

Hence, the 2-step transition probability from i to j is the i, j-entry of the matrix P^2.

More generally, it can be shown that $p_{ij}^{(n)} = (P^n)_{ij}$, as we have claimed before!